

5	(a)	(i)	Amount of work needed to take all its constituent nucleons apart so that they are separated an infinite distance from one another.	
		(ii)	Total binding energy of He-4 = $7.075175 \times 4 = 28.30070$ MeV Total binding energy of Be-9 = $6.462767 \times 9 = 58.16490$ MeV Total binding energy of C-12 = $7.675310 \times 12 = 92.10372$ MeV (2m, any one wrong will be awarded 1 m only for the question) Energy released = $92.10372 - (28.30070 + 58.16490) = 5.638120$ MeV	
	(b)	(i)	Nuclei/Atoms with same proton number/atomic number, but contain different numbers of neutrons/different atomic mass.	
		(ii)	1. mass = $238 \times 1.66 \times 10^{-27}$ = 3.95×10^{-25} kg 2. Volume = $4/3 \pi \times (8.9 \times 10^{-15})^3 = 2.95 \times 10^{-42}$ density = $(3.95 \times 10^{-25}) / (2.95 \times 10^{-42})$ = 1.3×10^{17} kg m⁻³	
		(iii)	The nucleus contains most of mass of atom. <i>either</i> The nuclear's diameter/volume is very much less than that of atom <i>or</i> The atom is mostly (empty) space.	

6	(a)		A photon is a quantum (packet) of electromagnetic radiation / wave / field	
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